Semester	T.E. Semester VI – Computer Engineering
Subject	Software Engineering
Subject In-charge	Prof. Divya Surve
Laboratory	Lab 313B

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Experiment Number	3	
Experiment Title	Cost Estimation Analysis of Project using Function Point (FP) and COCOMO Model	
CO Mapped		
Resources / Apparatus Required	Hardware: N/A	Software: Python 3 for numerical computation; IFPUG Function Point standard weight table; COCOMO-81 model (Basic, Intermediate, Detailed)
Description	**Aim:** To estimate the size, effort, development time and staffing of the **DocSense** project using Function Point (FP) Analysis and the three levels of the COCOMO model (Basic, Intermediate and Detailed), and to solve the given numerical problems on FP estimation and COCOMO. **Objectives:** 1. To identify the function point components (EI, EO, EQ, ILF and EIF) of DocSense from its Level 0 / Level 1 DFDs. 2. To calculate the Unadjusted Function Points (UFP) and the Adjusted Function Points (FP) using the 14 General System Characteristics. 3. To convert the Adjusted FP into program size (LOC and KLOC) and cross-check it against the independent KLOC estimate. 4. To calculate effort, development time and team size using Basic, Intermediate and Detailed COCOMO. 5. To solve the numerical problems on Basic, Intermediate and Detailed COCOMO and on FP analysis of a Library Management System DFD.	

Project Description:

DocSense is a document-understanding system built around the NuExtract3 vision-language model. Uploaded documents (PDF/files) are stored, processed by a fine-tuned NuExtract3 model and returned as structured Markdown/LaTeX output. The extracted content is indexed in a vector store so that a user can ask questions in a chat interface and receive answers through a Retrieval-Augmented Generation (RAG) pipeline that uses the Groq LLM service. The system also supports evaluation runs that produce a metrics report, and updating of the model version.

The Function Point counts were derived directly from the project's own Level 0 / Level 1 DFDs (Experiment 2): the external entities (End User, Evaluator, System Administrator), their processes and the three data stores (S3, DynamoDB, ChromaDB). The Adjusted FP is cross-checked against an independently estimated size in KLOC, and both feed into COCOMO.

The major functional modules identified from the DFD are:

- Document Upload and Storage (S3, DynamoDB)
- Structured Extraction using NuExtract3
- Chat and RAG Question Answering (ChromaDB, Groq LLM)
- Evaluation and Metrics Reporting
- Model Version Management

1. Function Point Estimation

FP analysis measures software size from its functional requirements, independent of the programming language. It counts five types of components – External Inputs (EI), External Outputs (EO), External Inquiries (EQ), Internal Logical Files (ILF) and External Interface Files (EIF).

Each is rated Low, Average or High and given the standard IFPUG weight:

Component	Low	Average	High
External Inputs (EI)	3	4	6
External Outputs (EO)	4	5	7
External Inquiries (EQ)	3	4	6
Internal Logical Files (ILF)	7	10	15
External Interface Files (EIF)	5	7	10

A. Components identified from the DocSense DFD

Type	Component	Complexity	Weight
EI	Document Upload (PDF/file)	Average	4
EI	Chat Question Submission	Low	3
EI	Run Evaluation Request	Low	3
EI	Model Version Update	Average	4
EO	Structured Markdown/LaTeX Output	High	7
EO	Metrics Report	Average	5
EQ	Chat Answer Retrieval	Average	4
ILF	D1: S3 – Raw Files	Low	7
ILF	D2: DynamoDB – File Metadata	Average	10

ILF	D3: ChromaDB – Vector Store	High	15
EIF	NuExtract3 Pretrained Weights	Low	5
EIF	Groq LLM Service	Low	5

Complexity follows the IFPUG matrices: transactions (EI/EO/EQ) by DETs and FTRs, files (ILF/EIF) by DETs and RETs. The structured Markdown/LaTeX output (EO) and the ChromaDB vector store (ILF) are the most complex items; both external interface files are Low.

2. UFP Calculation

Adding the weighted counts of all five categories:

Component Type	Count	Subtotal Weight
External Inputs (EI)	4	14
External Outputs (EO)	2	12
External Inquiries (EQ)	1	4
Internal Logical Files (ILF)	3	32
External Interface Files (EIF)	2	10
Unadjusted Function Points (UFP)	**—**	**72**

Therefore:

UFP = EI + EO + EQ + ILF + EIF

UFP = 14 + 12 + 4 + 32 + 10

UFP = 72

3. Complexity Adjustment (Value Adjustment Factor)

The adjusted function point count is given by:

FP = UFP × [0.65 + (0.01 × ΣFi)]

Each of the 14 General System Characteristics (GSC) is rated from 0 (no influence) to 5 (strong influence):

#	General System Characteristic	Rate	#	General System Characteristic	Rate
1	Data Communications	4	8	Online Update	2
2	Distributed Data Processing	3	9	Complex Processing	5
3	Performance	3	10	Reusability	2
4	Heavily Used Configuration	1	11	Installation Ease	2
5	Transaction Rate	1	12	Operational Ease	3
6	Online Data Entry	4	13	Multiple Sites	1
7	End-User Efficiency	3	14	Facilitate Change	3
Total Degree of Influence (ΣFi)					**37**

Therefore:

ΣFi = 37

VAF = 0.65 + (0.01 × 37) = 0.65 + 0.37 = 1.02

FP = 72 × 1.02

Adjusted FP = 73.44 FP

4. LOC Estimation

A backfire (FP-to-LOC) ratio converts FP into lines of code. For a Python/JavaScript project a blended ratio of about 55 LOC per FP is taken:

1 FP ≈ 55 LOC

Therefore:
LOC = FP × 55
LOC = 73.44 × 55
LOC = 4,039.2 ≈ 4,039 LOC

5. KLOC
COCOMO requires the size in KLOC.
KLOC = LOC / 1000
KLOC = 4039.2 / 1000
KLOC = 4.04 KLOC
The team had independently estimated the project at 4 KLOC from its modules (1.5 + 1.5 + 1.0 KLOC). The FP-derived 4.04 KLOC agrees to within about 1%, so the size used for COCOMO is:
Size = 4 KLOC

6. COCOMO Basic Model
Basic COCOMO estimates effort and schedule from the size alone. The three project types and their standard coefficients are:

Project Type	Nature of project	a	b	c	d
Organic	Small team, familiar domain, flexible constraints	2.4	1.05	2.5	0.38
Semi-detached	Mixed experience, medium size and complexity	3.0	1.12	2.5	0.35
Embedded	Large, tight hardware / real-time constraints	3.6	1.20	2.5	0.32

DocSense is a small-team project in a familiar problem domain with no rigid real-time constraints, so it is classified as an:
Organic Project (a = 2.4, b = 1.05, c = 2.5, d = 0.38)
Effort:
$E = a \times (KLOC)^b$
$E = 2.4 \times (4)^{1.05} = 2.4 \times 4.2871$
E = 10.29 Person-Months

7. Development Duration
Formula:
$D = c \times (E)^d$
Therefore:
$D = 2.5 \times (10.29)^{0.38} = 2.5 \times 2.4249$
D = 6.06 months

8. Staff Size
Formula:
Staff = Effort / Duration
Therefore:
Staff = 10.29 / 6.06
Staff = 1.70 ≈ 2 persons
The actual team has 4 members: COCOMO gives the effort a well-matched team needs, not a mandated team size; the extra members add schedule buffer and specialization (vision/adapter, decoder, data, evaluation).

9. Intermediate COCOMO

Intermediate COCOMO refines the nominal effort with cost driver attributes. The same size and mode are used, with 6 cost drivers rated for DocSense:

Cost Driver	Rating	Value	Cost Driver	Rating	Value
RELY (Reliability)	Nominal	1.00	STOR (Storage)	Nominal	1.00
CPLX (Complexity)	High	1.15	ACAP (Analyst Cap.)	Nominal	1.00
TIME (Exec. Time)	Nominal	1.00	LEXP (Language Exp.)	Low	1.07

Effort Adjustment Factor:

EAF = 1.00 × 1.15 × 1.00 × 1.00 × 1.00 × 1.07

EAF = 1.2305

The nominal effort of the Basic model (a = 2.4, b = 1.05, as followed in this experiment) is multiplied by the EAF:

E = 10.29 × 1.2305 = 12.66 Person-Months

D = 2.5 × $(12.66)^{0.38}$ = 2.5 × 2.6238 = 6.56 months

Staff = 12.66 / 6.56 = 1.93 ≈ 2 persons

The rise from Basic to Intermediate is driven almost entirely by CPLX (High, 1.15) and LEXP (Low, 1.07): VLM fine-tuning is genuinely complex, and the team has limited prior experience with the tools involved (PyTorch / HuggingFace fine-tuning, QLoRA).

10. Detailed COCOMO – Module-Level Breakdown

In Detailed COCOMO the cost drivers are applied per module instead of to the whole project. The 4 KLOC project is split into 3 modules, each with its own ACAP / TIME / STOR ratings reflecting the team's familiarity with that part of the system (organic coefficients applied per module):

Module	KLOC	ACAP	TIME	STOR
M1: NuExtract3 Integration & Fine-Tuning	1.5	1.19 (Low)	1.11 (High)	1.06 (High)
M2: RAG Pipeline Integration	1.5	1.00 (Nom.)	1.00 (Nom.)	1.00 (Nom.)
M3: Demo Interface & Evaluation Harness	1.0	0.86 (High)	1.00 (Nom.)	1.00 (Nom.)

Module effort = Nominal effort × module EAF, where Nominal = 2.4 × $(KLOC)^{1.05}$:

Module	Nominal (PM)	EAF	Effort (PM)
M1 (1.5 KLOC)	3.674	1.4002	5.144
M2 (1.5 KLOC)	3.674	1.0000	3.674
M3 (1.0 KLOC)	2.400	0.8600	2.064
Total Detailed Effort			**10.882**

Therefore:

Total Effort = 5.144 + 3.674 + 2.064

Total Effort = 10.88 Person-Months

D = 2.5 × $(10.88)^{0.38}$ = 6.19 months; Staff = 10.88 / 6.19 ≈ 1.76 ≈ 2 persons

Module 1 (NuExtract3 integration) alone carries a 1.40 EAF penalty from low ACAP, high TIME and high STOR ratings, while Module 3 (demo interface) costs less than its nominal effort (EAF 0.86) because it plays to the team's existing web-development strengths. This identifies Module 1 as the effort and risk hotspot, consistent with the debugging risk already flagged for Weeks 6–8 in the project's build plan.

11. Numerical Problems on FP Estimation and COCOMO

11.1 Basic COCOMO – 50 KLOC, Organic

Given: size = 50 KLOC, organic project, so a = 2.4, b = 1.05, c = 2.5, d = 0.38.

(a) Effort:

$E = 2.4 \times (50)^{1.05} = 2.4 \times 60.80 = 145.93$ Person-Months

(b) Development Time:

$D = 2.5 \times (145.93)^{0.38} = 2.5 \times 6.6431 = 16.61$ months

(c) Number of people required:

Staff = 145.93 / 16.61 = 8.79 ≈ 9 persons

11.2 Intermediate COCOMO – 200 KLOC, Semi-detached

Given: size = 200 KLOC, semi-detached (a = 3.0, b = 1.12). Development time uses the standard semi-detached constants c = 2.5, d = 0.35. Cost drivers:

Cost Driver	Rating	Value	Cost Driver	Rating	Value
RELY (Reliability)	High	1.15	STOR (Storage)	High	1.06
CPLX (Complexity)	High	1.15	ACAP (Analyst Cap.)	High	0.86
TIME (Exec. Time)	Nominal	1.00	LEXP (Language Exp.)	Low	1.07

(a) Effort Adjustment Factor:

EAF = 1.15 × 1.15 × 1.00 × 1.06 × 0.86 × 1.07

EAF = 1.28998 ≈ 1.29

(b) Effort:

Nominal $E = 3.0 \times (200)^{1.12} = 3.0 \times 377.71 = 1133.12$ PM

$E = 1133.12 \times 1.28998 = 1{,}461.70$ Person-Months

(c) Development Time:

$D = 2.5 \times (1{,}461.70)^{0.35} = 2.5 \times 12.8145 = 32.04$ months

(Average staffing = 1,461.70 / 32.04 ≈ 45.6 ≈ 46 persons.)

11.3 Detailed COCOMO – 128 KLOC, Embedded

Given: embedded mode, a = 3.6, b = 1.20. The module-level EAF (ACAP × TIME × STOR) is applied to the module-level nominal effort $3.6 \times (\text{KLOC})^{1.20}$, and the module efforts are summed.

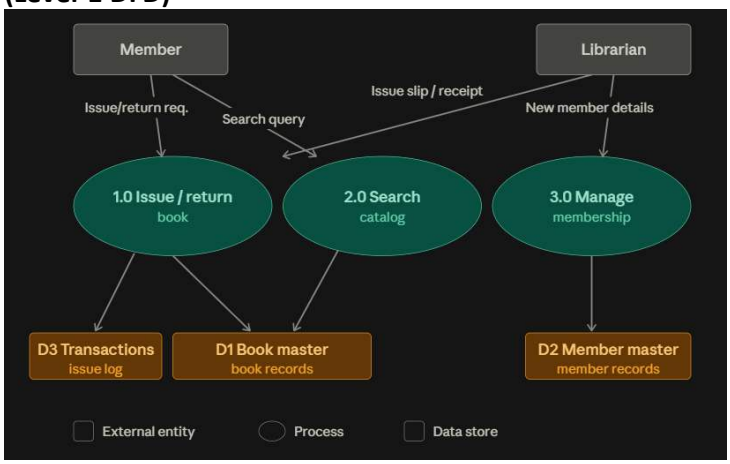
Module	KLOC	ACAP	TIME	STOR
Planning	32	1.19 (Low)	1.00 (Nom.)	1.00 (Nom.)
Design	46	1.00 (Nom.)	1.11 (High)	1.06 (High)
Implementation	50	0.86 (High)	1.30 (V.High)	1.21 (V.High)

Module	$\text{KLOC}^{1.20}$	Nominal (PM)	EAF	Effort (PM)
Planning	64.00	230.40	1.1900	274.176
Design	98.93	356.13	1.1766	419.026
Implementation	109.34	393.61	1.3528	532.468

Total Project Effort	**980.14**	**—**	**1,225.670**

Therefore:
Total Effort = 274.176 + 419.026 + 532.468
Total Effort = 1,225.67 Person-Months
For comparison, one 128 KLOC unit would need a nominal $3.6 \times (128)^{1.20}$ = 1,216.06 PM; the module-level nominal sum is 980.14 PM, raised by the cost drivers to 1,225.67 PM (average EAF ≈ 1.25). With c = 2.5, d = 0.32 the schedule is 24.33 months (about 50 persons).

11.4 Function Point Analysis – Library Management System (Level-1 DFD)



The DFD has two external entities (Member, Librarian), three processes (1.0 Issue/return book, 2.0 Search catalog, 3.0 Manage membership) and three data stores (D1 Book master, D2 Member master, D3 Transactions). The DFD does not give the DET/FTR/RET counts, so reasonable counts are assumed from the data flows (e.g. member ID, book ID, issue date, due date, return date, fine):

Type	Component	DET / FTR / RET	Complexity	Weight
EI	Issue/return request	8 DET, 2 FTR	Average	4
EI	Search query	3 DET, 1 FTR	Low	3
EI	New member details	8 DET, 1 FTR	Low	3
EO	Issue slip / fine receipt	8 DET, 2 FTR	Average	5
EQ	Search catalog result	6 DET, 1 FTR	Low	3
ILF	D1 Book master	10 DET, 1 RET	Low	7
ILF	D2 Member master	8 DET, 1 RET	Low	7
ILF	D3 Transactions	8 DET, 1 RET	Low	7
EIF	None in this scope	—	—	0

Therefore:
EI = 4 + 3 + 3 = 10; EO = 5; EQ = 3; ILF = 7 + 7 + 7 = 21; EIF = 0
UFP = 10 + 5 + 3 + 21 + 0 = 39
The 14 GSC ratings are not given, so they are assumed for a small, mostly online library system:

#	General System Characteristic	Rate	#	General System Characteristic	Rate
1	Data Communications	2	8	Online Update	3

2	Distributed Data Processing	0	9	Complex Processing	1
3	Performance	2	10	Reusability	1
4	Heavily Used Configuration	1	11	Installation Ease	2
5	Transaction Rate	3	12	Operational Ease	3
6	Online Data Entry	4	13	Multiple Sites	1
7	End-User Efficiency	3	14	Facilitate Change	2
Total Degree of Influence (ΣFi)					**28**

ΣFi = 28; VAF = 0.65 + (0.01 × 28) = 0.93
Adjusted FP = 39 × 0.93 = 36.27 ≈ 36 FP

12. Final Calculation Table

Summary of the DocSense estimates:

UFP	VAF	Adjusted FP	Size (KLOC)	KLOC used
72	1.02	73.44	4.04	4.0

Parameter	Basic	Intermediate	Detailed
EAF	1.00	1.2305	per module
Effort (PM)	10.29	12.66	10.88
Time (months)	6.06	6.56	6.19
Staff (persons)	1.70 ≈ 2	1.93 ≈ 2	1.76 ≈ 2

Summary of the numerical problems:

Problem	Result
11.1 Basic (50 KLOC)	E = 145.93 PM; D = 16.61 months; ≈ 9 persons
11.2 Intermediate (200 KLOC)	EAF = 1.29; E = 1,461.70 PM; D = 32.04 months
11.3 Detailed (128 KLOC)	Planning 274.18 + Design 419.03 + Implementation 532.47 = 1,225.67 PM
11.4 FP – Library System	UFP = 39; VAF = 0.93; Adjusted FP = 36.27

Program (wherever applicable)

```
# Experiment 3 - FP and COCOMO calculations (Python 3)
def cocomo(kloc, a, b, c, d, eaf=1.0):
    e = a * kloc ** b * eaf        # effort (person-months)
    t = c * e ** d                 # development time (months)
    return e, t, e / t             # staff = effort / time

def detailed(mods, a, b):          # mods = [(KLOC, EAF), ...]
    return [a * k ** b * f for k, f in mods]

# DocSense - function points
tdi = sum([4, 3, 3, 1, 1, 4, 3, 2, 5, 2, 2, 3, 1, 3])
afp = 72 * (0.65 + 0.01 * tdi)
print("FP   : UFP=72 TDI=%d AFP=%.2f KLOC=%.2f"
      % (tdi, afp, afp * 55 / 1000))

# DocSense - COCOMO (organic, 4 KLOC)
e, t, p = cocomo(4, 2.4, 1.05, 2.5, 0.38)
print("Basic: E=%.2f PM D=%.2f mo P=%.2f" % (e, t, p))
eaf = 1.15 * 1.07                  # CPLX x LEXP (others = 1.00)
e, t, p = cocomo(4, 2.4, 1.05, 2.5, 0.38, eaf)
print("Inter: EAF=%.4f E=%.2f PM D=%.2f mo" % (eaf, e, t))
m = detailed([(1.5, 1.19 * 1.11 * 1.06), (1.5, 1.0),
              (1.0, 0.86)], 2.4, 1.05)
print("Detail: %s total=%.2f PM"
      % ([round(x, 3) for x in m], sum(m)))

# Numerical problems
e, t, p = cocomo(50, 2.4, 1.05, 2.5, 0.38)
print("Q1   : E=%.2f PM D=%.2f mo P=%.2f" % (e, t, p))
eaf = 1.15 * 1.15 * 1.00 * 1.06 * 0.86 * 1.07
e, t, p = cocomo(200, 3.0, 1.12, 2.5, 0.35, eaf)
print("Q2   : EAF=%.4f E=%.2f PM D=%.2f mo" % (eaf, e, t))
m = detailed([(32, 1.19), (46, 1.11 * 1.06),
```

	(50, 0.86 * 1.30 * 1.21)], 3.6, 1.2) print("Q3 : %s total=%.2f PM" % ([round(x, 2) for x in m], sum(m))) ufp, vaf = 10 + 5 + 3 + 21, 0.65 + 0.01 * 28 print("Q4 : UFP=%d VAF=%.2f AFP=%.2f" % (ufp, vaf, ufp * vaf))
Output (wherever applicable)	FP : UFP=72 TDI=37 AFP=73.44 KLOC=4.04 Basic: E=10.29 PM D=6.06 mo P=1.70 Inter: EAF=1.2305 E=12.66 PM D=6.56 mo Detail: [5.144, 3.674, 2.064] total=10.88 PM Q1 : E=145.93 PM D=16.61 mo P=8.79 Q2 : EAF=1.2900 E=1461.70 PM D=32.04 mo Q3 : [274.18, 419.03, 532.47] total=1225.67 PM Q4 : UFP=39 VAF=0.93 AFP=36.27
Conclusion:	The cost estimation of DocSense was performed using Function Point Analysis and the Basic, Intermediate and Detailed COCOMO models. The count of 72 UFP, adjusted by a VAF of 1.02, gives 73.44 FP; with a backfire ratio of about 55 LOC per FP this is approximately 4.04 KLOC, nearly identical to the 4 KLOC estimated independently from the module breakdown. The two estimates therefore validate each other. Basic COCOMO gives 10.29 person-months over 6.06 months (about 2 people, fewer than the team's actual 4, since COCOMO estimates the effort a well-matched team needs and not a mandated team size). Intermediate COCOMO raises the effort to 12.66 person-months, driven by product complexity (High) and language experience (Low). Detailed COCOMO gives 10.88 person-months and identifies Module 1 (NuExtract3 integration, EAF 1.40) as the effort and risk hotspot, while Module 3 (EAF 0.86) costs less than nominal. The numerical problems showed how the models scale: 50 KLOC organic needs 145.93 PM over 16.61 months (about 9 people); 200 KLOC semi-detached with EAF 1.29 needs 1,461.70 PM over 32.04 months; and 128 KLOC embedded, estimated module by module, needs 1,225.67 PM. Effort grows faster than size because $b > 1$ – the 50 KLOC project is 12.5 times the size of DocSense but needs about 14.2 times the effort. The Library Management System DFD gave 39 UFP and 36.27 adjusted FP.